

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459449.5611 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.029
Transit depth (Rp/Rs)^2	2.57 +/- 0.94 [%]
Orbital Inclination (inc)	89.26 +/- 1.9
Airmass coefficient 1 (a1)	1.091 +/- 0.016
Airmass coefficient 2 (a2)	-0.063 +/- 0.053
Scatter in the residuals of the lightcurve fit is	1.43 %
Best Comparison Star	#2 - [401, 215]
Optimal Aperture	6.16
Optimal Annulus	21.76
Transit Duration (day)	0.0267 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.16 $\pm$ 0.029 vs 0.1326 (deviation 0.94 σ)
Duration fit vs published	0.0267 d vs 0.0737 d (deviation 7.84 σ)
Mid-time O-C	-3083.6 min
Depth SNR	5.5
Residual scatter	1.43 %

Light curve

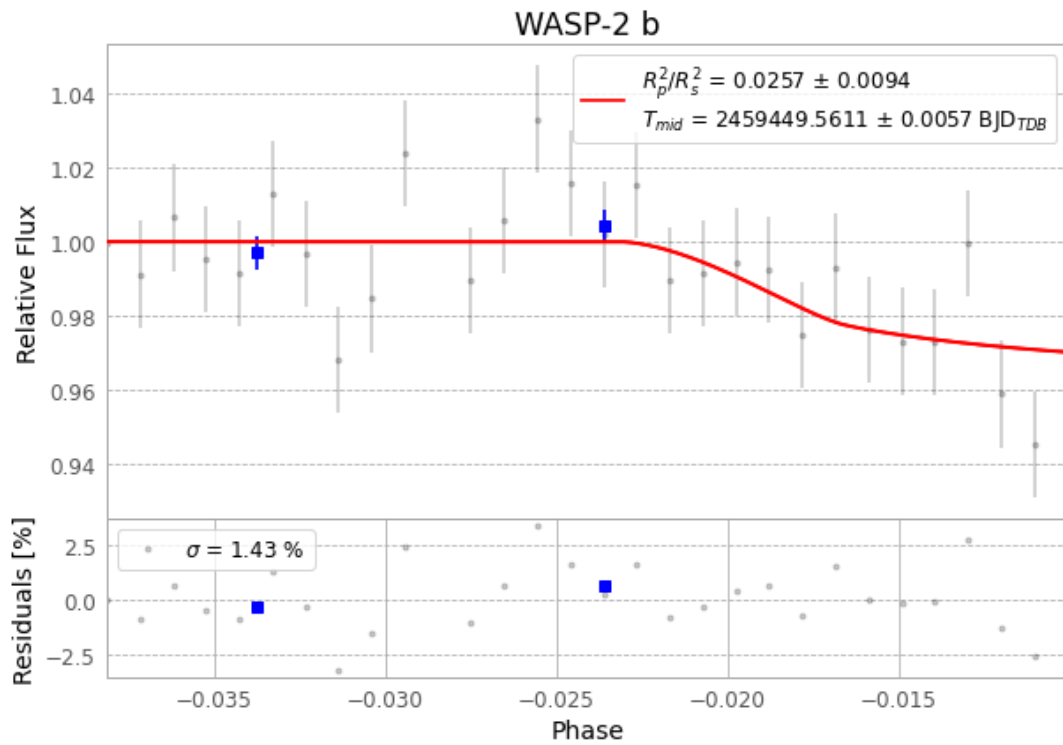


Figure 1 — FinalLightCurve_WASP-2 b_2021-08-24.png (SHA-256 f9fbbc082437b77e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [359, 98], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cf51d2c758e11d16...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

30 FITS frames (20 MB), WASP-2210825030012.FITS → WASP-2210825042712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-210825.log (da76ad3a7dfa...), FinalLightCurve_WASP-2 b_2021-08-24.png (f9fbbc082437...), FinalLightCurve_WASP-2 b_2021-08-24.csv (b12dfa0b89ec...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 21:54Z.